

Predicting stock price changes under strong concept drift

[Progress Report]

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ABSTRACT

Prediction tasks in certain real-world domains are faced with the problem that some underlying variables may change over time. This phenomena is defined as concept drift. Due to this phenomena, models tend to give less accurate results, since previous data that the model has been trained on does not reflect the current data.

This report investigates the use of incorporating concept drift detectors with machine learning models in order to identify when a concept drift has occurred. With this knowledge the model can then react accordingly. To test the proposed method two different real world financial data-sets are used and the models used are three different variants of a long short-term memory neural network.

Keywords

Concept Drift, Fintech, Machine Learning, Time Series Prediction

1. INTRODUCTION

Time series prediction is the process of forecasting future values from a temporally ordered sequence of data [1]. Time series prediction is used in multiple real-world domains, including economic forecasting [2], sales forecasting [3] and network load forecasting [4]. This report aims to investigate time series prediction in the finance domain, specifically the stock market domain.

The stock market (also known as the equity market) [5] is essentially an organized and regulated financial market where tradable financial assets, referred to as securities, can be purchased. Some examples of securities include stocks, bonds and shares [6]. Applying time series prediction to the stock market is no trivial task. This is mainly due to the efficient market hypothesis [7], which states that the current asset price reflects any and all available information. This implies that all of the new information that enters the market is used, hence making it very difficult to predict future prices based off of old prices. Due to this dynamic nature of the stock market, models that have been trained on past, and no longer valid data start to give less accurate results.

2. AIMS AND OBJECTIVES

The main research question that this report aims to answer is; “Can a machine learning model that has been trained

on a finance data-set get better results if it undergoes a retraining process every time a concept drift has been detected?” This research question is going to be answered by completing the following objectives:

- Objective 1 (O1): Compare and contrast state-of-the-art concept drift detectors in order to find the most suitable concept drift detector.
- Objective 2 (O2): Compare and improve upon vanilla machine learning models by incorporating the models with the concept drift detector selected in O1.

In O1 a number of state-of-the-art concept drift detectors that are known to give good results will be tested on two different finance data-sets.

In O2 an instance selection approach [8,9] is going to be used. Three similar machine learning models that are known to make good predictions on stock prices are going to be tested on two finance data-sets [10–12]. Each model will then be fitted with a concept drift detector that has been chosen from O1. Each time this concept drift detector detects a concept drift, the model will under go a retraining process. During the evaluation process, the results of the vanilla models will be compared to the results of the models that were fitted with a concept drift detector. If the models with concept drift detectors give better results then the vanilla models, then this would suggest that the answer to the above research question is indeed a yes.

3. LITERATURE REVIEW

The market has been previously described as a random walk model [13,14]. If this was the case then it would practically be impossible to make any sort of real accurate predictions based off of previous prices. It would also mean the most accurate possible prediction for any future price is the current price. There is however a considerable amount of evidence to suggest that this is not the case [15]. A survey conducted in 2020 found that a large number of machine learning techniques are used as analytical tools by traders in order to make predictions on different financial instruments [5]. Some of these various techniques include k-means clustering, neural networks and genetic algorithms. This report focuses on further enhancing one of these techniques, specifically neural networks, by using concept drift detectors. Neural networks were chosen as the preferred technique as they already give good results and are frequently used to predict stock prices [5].

The stock market is dynamic and is constantly fluctuating. Due to this constant fluctuation certain underlying hidden concepts start to change over time. This phenomena is classified as concept drift. When concept drift occurs machine learning models tend to give substantially less accurate results. This is due to the fact that their training data's concepts are no longer inline with the current data's concepts. A very clear example of this can be seen in models that make predictions of the temperature on a given day [16]. In this scenario, one important concept is that of the seasons, and how when a change in season occurs, it also brings along a concept drift. There are multiple types of concept drift [8], and to better understand each type of drift, examples are given based of the temperature prediction model example.

- Sudden → This kind of concept drift happens very rapidly and over a short amount of time (example: heat waves).
- Gradual → This kind of concept drift happens over a prolonged period of time (example: global warming).
- Recurring → This kind of concept drift follows a cyclic pattern, where the given concept drifts shift between a select set of concepts (example: seasons of the year).

From this understanding of concept drift a hypothesis can be made of what the ideal effective learner (referring to some machine learning algorithm) should look like. It should be able to quickly adapt to concept drift, while still being robust enough to noise [8]. It should also save concepts so that their data can be used later by the model if they ever reappear [8]. A number of different techniques have been implemented to try and tackle concept drift. Most of these techniques can be clustered into one of the following categories [8,9]:

- Instance selection → These types of techniques select instances that are either similar to the current concept, or else they only select recent instances. An example of such an algorithm is the *window resize* algorithm for batch data [17].
- Ensemble learning → These type of techniques maintain a set of concept descriptions, and through some defined heuristic, the most similar concept is selected. An example of this approach can be seen in the *conceptual clustering and prediction* framework [18].
- Instance weighting → These type of techniques use multiple concepts at a time, but give more weighting to the concepts that are similar to the current concept. When implementing such a model, one needs to be careful not to overfit it. An example of such an algorithm is concept drift 3 (CD3) [19].

All of the above techniques rely on concept drift detectors to know when a change in concept has occurred. Put simply, a concept drift detector keeps track of the current concept, and when it detects with a high level of certainty that a concept drift has occurred, it returns an alert. Most concept drift detectors can be clustered in one of the following methods [20]:

- Sequential Analysis based methods: These methods evaluate and keep track of how well the current model

is doing, and when the model's performance dips under a certain predefined threshold, they make an alert that a concept drift has occurred.

- Statistical based methods: These methods make use of statistical techniques, such as standard deviation to detect concept drifts,
- Windows based methods: These methods contain a fixed reference window alongside a sliding window. The sizes of these windows can either be static or dynamic depending on the implementation. If there is a drastic difference in the distribution between the sliding window and the reference window then the concept drift detector returns an alert.

4. METHODOLOGY

4.1 Selected Machine Learning Models

When it came to choosing the machine learning models, a number of options were originally drafted. Ideally, the chosen models follow as closely as possible the previously defined description of an ideal learner, located in section 3. It would also be ideal if the machine learning models have already been previously tested on finance data-sets in past literature. The reason behind this being that the results generated by the models without the concept drift detectors can be easily compared to the models that have a concept drift detectors attached to them. Due to the above reasons the following models were chosen:

- A vanilla Long Short-Term Memory (LSTM) neural network [10,21]
- An Attentive Long Short-Term Memory (ALSTM) neural network [10,22]
- An Adversarial Attentive Long Short-Term Memory (Adv-ALSTM) neural network [10]

These three models were all implemented and applied in [10], which aims to enhance stock movement prediction through the use of adversarial training. The original code for these three models can be found here:

<https://github.com/fulifeng/Adv-ALSTM>

4.1.1 LSTM neural networks explained

A LSTM neural network [21] is a type of recurrent neural network [23] that is capable of learning long-term dependencies. This means that a LSTM neural network is capable of remembering data which is deemed important for a long period of time. Through this the model is capable of making a connection with past data to the present data, hence partially solving the problem of long-term dependencies. In [10], the LSTM is tasked with predicting if the given stock closing price is going to increase or decrease in the next time-step.

4.1.2 ALSTM neural networks explained

As seen in [10,22], an ALSTM neural network further builds upon the previous LSTM neural network. This is mainly accomplished by the addition of a temporal attention layer. This layer compresses data at different time-steps into an overall representation with adaptive weights. The aim of this layer is to use multiple compressed representations of data at the same time, giving more importance to those with

a higher weighting. Hence, because of this, the temporal attention layer can be considered as an instance weighting technique.

4.1.3 Adv-ALSTM neural networks explained

As seen in [10], an Adv-ALSTM further enhances the previously explained ALSTM neural network. This is accomplished through adversarial training [24, 25]. Adversarial training is the process of adding malicious data to the training data. This malicious data is generated by systematically copying and then transforming parts of training data. Through this method not only does the model have more data to work with but it also becomes more robust to noise.

4.2 Selected Data-sets

For the three previously mentioned neural networks to work they require a valid data-set. A stock market data-set consists of a list of temporally ordered data points. For this use case, each data point should represent a specific day. It also needs to be ensured that all data points that are weekends and public holidays are removed, as the stock market is not open on these days [26].

After some consideration the following two data-sets were selected; the stocknet data-set [11] and another unnamed data-set that has been used in [12] to predict stock prices. This unnamed data-set shall be referred to as KDD17.

4.2.1 Stocknet

This data-set contains historical data of 88 different stocks from the 1st of January 2014 till the 1st of January 2016, with a total of 652 data points for each stock, which gives a total number of 57,376 different data points. It contains stocks from both the National Association of Securities Dealer Automated Quotation market and the New York stock exchange market. All of it's data was collected from *Yahoo! Finance*.

4.2.2 KDD17

This data-set contains historical data of 50 different stocks from the 1st of January 2006 till the 1st of January 2016, with a total of 2518 data points for each stock, which gives a total number of 125,900 data points. All of it's stocks belong to the United States markets. Once again, all of the data was collected from *Yahoo! Finance*.

4.3 Objective 1

The aim of objective 1 is to find the most suitable concept drift detector. To do this an experiment is going to be carried out, where a number of concept drift detectors are going to be tested and evaluated on the stocknet and KDD17 data-sets. The ideal concept drift detector should be able to both distinguish concept drift from noise and detect both gradual and sudden concept drift. The four following concept drift detectors were chosen to be tested:

- Early Drift Detection Method (EDDM) [27]
- Drift Detection Method based on Hoeffding's bounds with moving average-test (HDDM_A) [28]
- Drift Detection Method based on Hoeffding's bounds with moving weighted average-test (HDDM_W) [28]

- Page-Hinkley method for concept drift detection (PH) [29]

These four specific concept drift detectors were chosen over other detectors as not only do they each give good to great results [20], but together they also cover the three methods mentioned in Section 3, where:

- EDDM takes a statistical based approach.
- HDDM_A and HDDM_W take a windows based approach.
- PH takes a sequential analysis based approach.

4.4 Objective 2

As explained in section 3, the report will aim to partially solve the problem of concept drift by retraining the models every time a concept drift has occurred. In this case, retraining is defined as feeding more training data to the model. Algorithm 1 formalises this proposed method.

Algorithm 1:

```

1 Initialize: machine learning model:  $M$ 
2 Initialize: concept drift detector:  $D$ 
3 Initialize: list with potentially infinite length
   containing a stream of real values:  $data$ 
4 Initialize: empty list:  $currentConcept$ 
5 Split  $data$  into  $trainingData$ ,  $validationData$  and
    $testingData$ 
6 Train model  $M$  on  $trainingData$ 
7 Validate model  $M$  on  $validationData$ 
8 foreach element  $i \in testingData$  do
9   Add element  $i$  to concept detector  $C$ 
10  Append element  $i$  to list  $currentConcept$ 
11  if  $D.driftDetected$  then
12    if  $currentConcept.length \geq 63$  then
13      Update model  $M$  using  $currentConcept$  as
        training data
14    else
15      Train model  $M$  on the last 63 datapoints.
16    end
17  Clear  $currentConcept$  list
18 end
```

Originally the models were going to be retrained on the previously observed concept, but during the tests ran in O1, it was observed that certain concepts were very short, and hence these short concepts did not have enough data for the model to update properly. To solve this problem a threshold system was setup (refer to line 12 in Algorithm 1). This threshold system ensures that the models will resume training on at least 3 months of data, meaning a total of 63 data points, as the stock market is open an average of 21 days per month [26]. This number is prone to change as more tests are conducted.

4.5 Evaluation Plan

To evaluate Algorithm 1 the Accuracy (ACC) [30] and Matthews Correlation Coefficient (MCC) [31] metrics are going to be used. As development of the proposed method continues, more evaluation metrics might be added.

Accuracy represents the proportion of samples that are correctly classified. Accuracy can be described as the percentage of correctly made predictions.

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Total number of predictions}} \\ = \frac{TP + TN}{TP + TN + FP + FN}$$

where:

- Accuracy $\in [0, 1]$ (The bigger the better)
- TP = True positives
- TN = True negatives
- FP = False positives
- FN = False negatives

While accuracy is a simple and elegant metric, it does not take into account class imbalance. So to combat this, the MCC metric will also be used.

MCC treats the true values and the predicted values as binary variables, and computes their phi-coefficient [31].

$$\text{MCC} = \frac{TP * TN - FP * FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$$

where:

- MCC $\in [-1, 1]$ (The bigger the better)

These metrics were chosen over other metrics as the results can easily be compared to the results generated in [10].

5. RESULTS

5.1 Objective 1

As stated in section 4.3, O1 entails that four different concept drift detectors be tested on the stocknet and KDD17 data-sets. This experiment has been conducted and it yielded the following results:

Concept Drift Detector	No. of concepts in stocknet	No. of concepts in KDD17	Total no. of concepts
EDDM	408	378	786
HDDM _A	6383	6624	13007
HDDM _w	3128	2588	5716
PH	1058	404	1462

Table 1: The different detectors tested on both data-sets

As can be seen by Table 1, HDDM_A detected the most concept drifts. This was expected as the same has been seen in [20, 32].

One also needs to keep in mind type I (false positive) errors, as even though it is true that HDDM_A detected the most concept drifts, it is not known how many of those concept drifts have actually occurred. However, as seen in [20, 28, 32], HDDM_A is quite resilient to these kinds of errors. Infact, as seen in [32], which is a survey conducted in 2018 that compared 15 different concept drift detectors showed that HDDM_A gave the best results, in terms of precision, recall and MCC. For these two reasons HDDM_A was chosen as the concept drift detector to be incorporated with the machine learning models.

5.2 Preparing For Objective 2

In order for objective 2 to be achieved, the following objectives need to first be tackled:

- Objective 2.1: Replicate results shown in [10]
- Objective 2.2: Find the optimal parameter values for the HDDM_A concept drift detector

5.2.1 Objective 2.1

The aim of objective 2.1 is to see if the results found in [10] can be successfully replicated. This objective has to be achieved in order to fairly compare the vanilla model's results, with those that will be generated by Algorithm 1. The three models mentioned in section 4.1 have been implemented, and their results compared to the results given in [10]:

Model Name	Paper ACC(%)	Machine ACC(%)	ACC - Percentage Difference
LSTM	53.18 (± 5)	53.33 (± 1)	-0.282060925
ALSTM	54.9 (± 0.7)	55.26 (± 1)	-0.655737705
Adv-ALSTM	57.2	57.2 (± 008)	0

Table 2: ACC results of models tested on the stocknet data-set

Model Name	Paper MCC	Machine MCC	MCC - Percentage Difference
LSTM	0.0674 (± 0.005)	0.0733 (± 0.3)	-8.753709199
ALSTM	0.1043 (± 0.07)	0.1041 (± 0.2)	0.191754554
Adv-ALSTM	0.1483	0.1552 (± 0.2)	-4.652730951

Table 3: MCC results of models tested on the stocknet data-set

Model Name	Paper ACC(%)	Machine ACC(%)	ACC - Percentage Difference
LSTM	51.62 (± 0.4)	51.33 (± 0.3)	0.561797753
ALSTM	51.94 (± 0.7)	51.61 (± 0.4)	0.635348479
Adv-ALSTM	53.05	53.24 (± 0.5)	-0.358152686

Table 4: ACC results of models tested on the KDD17 data-set

Model Name	Paper MCC	Machine MCC	MCC - Percentage Difference
LSTM	0.0183 (± 0.006)	0.0184 (± 0.005)	-0.546448087
ALSTM	0.0261 (± 0.01)	0.02607 (± 0.009)	0.114942529
Adv-ALSTM	0.0523	0.0526 (± 0.009)	-0.573613767

Table 5: MCC results of models tested on the KDD17 data-set

Tables 2 and 3 show the ACC score and the MCC score of the stocknet data-set, while tables 4 and 5 show the ACC score and MCC score of the KDD17 data-set. The Paper ACC and Paper MCC columns refer to the results that were obtained in [10], while the Machine ACC and Machine MCC columns refer to the replicated results. As can be seen from tables 2-5, the replicated results have minimal differences when compared to [10], with the majority having less than one percent difference.

As expected, the Adv-ALSTM model consistently generated the best results, while the LSTM model generated the worst results. From these results it can be concluded that the results found in [10] have been successfully replicated.

5.2.2 Objective 2.2

The aim of this objective is to evaluate which is the optimal parameter values for the HDDM_A concept drift. The most important parameter that HDDM_A makes use of is the drift confidence parameter. This parameter controls how high the probability of a drift occurring needs to be in order

for the concept drift detector to return an alert. By default this is set to 99%. Different values of the drift confidence parameter were tested.

Drift confidence	Average number of concept drifts per stock	Total number of drifts
0.999	73	6383
0.99	83	7294
0.9	102	8924
0.8	111	9731

Table 6: HDDM_A tested on the stocknet data-set

Drift confidence	Average number of concept drifts per stock	Total number of drifts
0.999	130	6624
0.99	149	7580
0.9	185	9438
0.8	203	10329

Table 7: HDDM_A tested on the KDD17 data-set

As was expected, the lower the drift confidence parameter the more concepts are detected. Since it takes time for the models to retrain, it was decided to keep the default value of 0.99. Any higher then the default value and not enough concepts would be detected, any lower and the models would retrain to often. In order to better visualise the concept drift detector, the following graphs were generated:

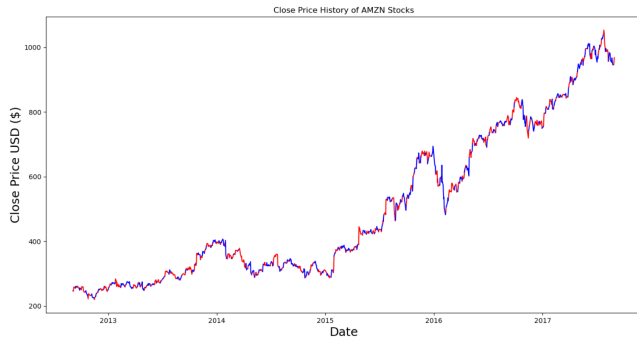


Figure 1: HDDM_A tested on Amazon stocks

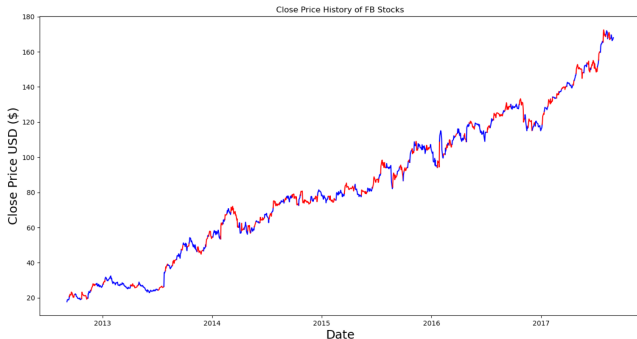


Figure 2: HDDM_A tested on Facebook stocks



Figure 3: HDDM_A tested on Google stocks

Figures 1-3 show HDDM_A applied to some of the stocks found in the stocknet data-set. A change in colour signify that a concept drift has happened.

6. CONCLUSION

The initial phases of development for Algorithm 1 has already started. If time permits, another interesting experiment that can be carried out, is to check if the proposed method can be applied to models that have been trained on non-finance data-sets, in order to see if some sort of generalisation can be made.

All of the code will be written in Python3. The code used to generate the results can be found here: <https://github.com/tm-26/Enhancing-stock-price-prediction-models-by-using-concept-drift-detectors>

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